

Non-Graded Project: End-to-End Embedding System Validation for GlobalRetail

Description

Transform GlobalRetail's failing keyword search system into a production-ready embedding-based solution by applying all course skills: building and validating semantic search with recall metrics, diagnosing data quality issues through UMAP visualization, benchmarking multiple models across cost-accuracy-latency trade-offs, and delivering an executive recommendation for a \$2M infrastructure investment.

Learning Objectives

By the end of this project, learners will be able to:

- Construct complete embedding-based search systems integrating sentence-transformers, FAISS, and recall validation. (LO1)
- Apply UMAP visualization and DBSCAN clustering to systematically diagnose data quality issues affecting embedding performance. (LO2)



- Execute comprehensive multi-model benchmarking across accuracy, latency, and cost dimensions with weighted decision frameworks. (LO3)
- Synthesize all course learnings into professional, deployable search infrastructure serving 500M monthly queries.
- Evaluate architectural trade-offs and deliver executive-ready recommendations to balance business priorities with technical constraints.



Project Components

Introductory Slide

Title: Lead ML Engineer Mission: Transform GlobalRetail's Search or Lose \$90M Annually

Subtitle: 60 minutes to validate an embedding system that will inform a \$2M infrastructure investment

Tools to be used

Python 3.9+ with pandas, NumPy, and matplotlib

sentence-transformers library (multiple models: all-mpnet-base-v2, all-MiniLM-L6-v2, gte-small)

FAISS for vector indexing and similarity search

UMAP for dimensionality reduction and visualization

DBSCAN from scikit-learn for outlier detection

Jupyter Notebook or Python IDE for development

Excel or Google Sheets for cost modeling and decision matrices

Git for version control

Scenario

You are the lead machine learning engineer at GlobalRetail, the world's third-largest e-commerce platform with 50 million products across 20 countries. GlobalRetail's current keyword-based search system delivers only 61% user satisfaction, significantly below competitors like Amazon (78%) and eBay (74%). Users complain about irrelevant results, especially for complex queries like "waterproof bluetooth headphones for running." The CEO has made search improvements a top 3 company priority, with a goal to reach 75% satisfaction within 6 months.

Your SVP of Engineering has tasked you with building and validating a production-ready embedding-based search system as the replacement for the legacy keyword system. This project will inform a \$2M infrastructure investment decision and affect 500 million monthly searches. You have 60 minutes to deliver a comprehensive analysis covering system implementation, data quality diagnosis, multi-model evaluation, and an executive recommendation.

Instructions for learners

Step 1: Build Baseline Embedding System

Download the GlobalRetail product catalog dataset (sample: 10,000 products with product_id, title, description, category, price).

Clean and prepare data: handle missing descriptions, combine title + description fields, remove duplicates.

Select initial embedding model: all-MiniLM-L6-v2 (384 dimensions) as baseline.

Generate embeddings for all products using sentence-transformers.

Build FAISS IndexFlatL2 index and add product embeddings.

Download test query set (100 queries with ground truth relevance labels).

Implement recall@5 and recall@10 calculation: for each query, retrieve top-10 results and compare against ground truth.

Calculate average recall@5 and recall@10 across all test queries.

Document baseline performance: if recall@10 improves from keyword baseline (51%) to embedding baseline (65%), you've achieved 14 percentage point improvement.

Step 2: Data Quality Analysis with UMAP and DBSCAN

Generate UMAP 2D visualization of product embeddings with `n_neighbors=15`, `min_dist=0.1`.

Create scatter plot color-coded by product category to assess cluster quality.

Identify categories with poor clustering (scattered points, overlap with other categories).

Apply DBSCAN (`eps=0.5`, `min_samples=5`) to UMAP coordinates to detect outliers automatically.

Extract outlier product_ids (cluster label = -1) and analyze their metadata.

Investigate patterns: sample 30-50 outliers and check for missing descriptions, very short text (<50 words), specific problematic vendors, mislabeled categories.

Formulate 3+ testable hypotheses with quantitative evidence:

Example: "Products from Vendor X account for 35% of outliers due to poor metadata quality."

Example: "Products with descriptions <75 words have 3x higher outlier rate."



Example: "Electronics category shows high variance due to mixing mobile phones and computer accessories."

Propose prioritized cleanup plan: fix high-impact issues first (e.g., Vendor X products, minimum description length filter).

Step 3: Multi-Model Benchmarking

Select 3 models for comprehensive evaluation:

- all-mpnet-base-v2 (768 dimensions, high accuracy, slower).
- all-MiniLM-L6-v2 (384 dimensions, baseline).
- gte-small (384 dimensions, optimized for speed).

ACCURACY BENCHMARKING:

- Generate embeddings for test corpus using each model.
- Build FAISS index for each model.
- Calculate recall@5, recall@10, and MRR for each model on the 100-query test set.

LATENCY BENCHMARKING:

- Measure query encoding time for 100 test queries (calculate p50, p95, p99 percentiles).
- Measure FAISS search time separately.
- Calculate total latency = encoding + search time.

COST ANALYSIS:

→ One-time costs: embedding generation for 50M products (compute hours × EC2 pricing).

→ Monthly recurring costs: query serving compute, vector storage (S3/EBS), data transfer.

→ Project costs at 500M queries/month scale.

Build comprehensive comparison table with all metrics.

Step 4: Production Architecture Design

Design scalable architecture for 50M products and 500M monthly queries:

- Vector index sharding strategy: partition index across multiple servers.
- Caching layer: Redis for frequently searched queries (estimate 30% cache hit rate).
- Load balancing: distribute query traffic across multiple embedding servers.
- Auto-scaling: configure EC2 auto-scaling groups based on query volume.

Specify infrastructure requirements:

- Compute: number and type of EC2 instances for query serving.
- Storage: vector index size ($50\text{M products} \times \text{dimension} \times 4 \text{ bytes per float}$).
- Network: bandwidth requirements for 500M queries/month.

Define monitoring and alerting:

→ Metrics: p95/p99 latency, recall@10 on canary queries, cache hit rate, error rate.

→ Alerts: latency >100ms p95, recall <60%, error rate >1%.

→ Dashboard: real-time visualization of key performance indicators.

Plan gradual rollout strategy: 5% → 25% → 50% → 100% traffic over 4-6 weeks.

Step 5: Build Weighted Decision Matrix

Define decision criteria with business-justified weights:

- Accuracy (35%): Recall improvement directly impacts revenue.
- Latency (30%): Must stay under 100ms for acceptable user experience.
- Cost (25%): Annual operating budget capped at \$800K.
- Implementation Complexity (10%): Time to production and maintenance burden.

Score each model on 1-10 scale for each criterion with clear rubric:

- Accuracy: 10 = recall@10 >70%, 5 = recall@10 60-65%, 1 = recall@10 <55%.
- Latency: 10 = p95 <50ms, 5 = p95 75-100ms, 1 = p95 >150ms.
- Cost: 10 = <\$500K/year, 5 = \$600-700K/year, 1 = >\$1M/year.

Calculate weighted scores: $\text{total_score} = \sum(\text{criterion_score} \times \text{weight})$.

Rank models by total weighted score.

Document trade-offs: if Model A wins, explain why its strengths outweigh its weaknesses for GlobalRetail's specific needs.

Step 6: Executive Recommendation and ROI Analysis

Select recommended model based on decision matrix.

Quantify business impact:

→ Recall improvement: from 51% (keyword) to 68% (embedding) = +17 percentage points.

→ Estimated satisfaction improvement: from 61% to 75%+ = +14 percentage points.

→ Revenue impact: 14% satisfaction improvement $\times$ \$15M per 1% = \$210M annual revenue protection.



ROI calculation: $(\$210\text{M revenue gain} - \$400\text{K additional annual cost}) / \$2\text{M initial investment} = 10,480\%$ ROI.

Payback period: $\$2\text{M investment} / (\$210\text{M annual benefit} - \$400\text{K cost}) \approx 0.01$ years = 4 days.

Document risks and mitigation:

- Risk: Embeddings may not capture domain-specific terminology → Mitigation: Fine-tune model on GlobalRetail data.
- Risk: Higher infrastructure costs → Mitigation: Implement aggressive caching (30%+ hit rate).
- Risk: Migration complexity → Mitigation: Gradual rollout with immediate rollback capability.

Prepare 2-3 slide executive summary highlighting key recommendation, business impact, and next steps.

Compile Your Findings

1. Requirement 1: Production-Quality Python Codebase



A complete implementation integrating all five project phases into a cohesive system.

Include: Data preparation and cleaning scripts, embedding generation with sentence-transformers, FAISS index construction, recall validation logic, UMAP visualization code, DBSCAN outlier detection, multi-model benchmarking framework, and clear documentation connecting each component.

2. Requirement 2: Data Quality Analysis Report

A comprehensive diagnosis of data quality issues affecting embedding performance.

Include: UMAP visualizations color-coded by category, DBSCAN outlier detection results with counts and percentages, metadata pattern analysis for sampled outliers, three prioritized hypotheses with quantitative supporting evidence, and proposed cleanup workflow.

3. Requirement 3: Model Comparison Matrix and Decision Framework

A structured evaluation of all three embedding models with a weighted recommendation.

Include: Accuracy benchmarks (recall@5, recall@10, MRR), latency measurements (p50, p95, p99), cost projections at 500M queries/month scale, weighted decision matrix with justified criteria weights, and final model ranking with trade-off analysis.

4. Requirement 4: Executive Presentation and ROI Analysis

A board-ready recommendation for Global Retail's C-suite on the infrastructure investment.

Include: Recommended model selection with supporting rationale, quantified business impact (satisfaction improvement, revenue protection calculations), ROI analysis with payback period, identified risks with mitigation strategies, and phased rollout plan.

5. Requirement 5: Monitoring Dashboard Specification

A production monitoring strategy ensuring sustained system quality at scale.

Include: Key performance indicators (p95/p99 latency, recall@10, cache hit rate, error rate), alert thresholds with specific trigger values, dashboard wireframes or mockups, and rollback decision criteria with escalation procedures.

Summary



This project provides hands-on experience with end-to-end embedding system development through practical application of semantic search construction, data quality diagnostics, multi-dimensional model evaluation, and production architecture design. Participants have developed competency in building FAISS-based search systems with recall validation, applying UMAP and DBSCAN for systematic quality diagnosis, executing comprehensive model benchmarking with weighted decision frameworks, and translating technical analysis into executive recommendations with ROI justification, which are directly applicable to leading ML infrastructure initiatives at enterprise scale.

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review



Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.